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# The mean flow, velocity dispersion, energy transfer and evolution of rotating and growing dark matter haloes

Zhijie (Jay) Xu,[1]⋆
[1]*Physical and Computational Sciences Directorate, Pacific Northwest National Laboratory; Richland, WA 99352, USA*

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**ABSTRACT**

Where does the angular momentum of a dark matter halo come from, why is the spin parameter close to 0.035 for haloes of all masses, and how is the angular momentum distributed inside a halo? We address these questions with the moment (Jeans) equations of an axisymmetric halo that rotates and grows at the same time. Three exact budget laws follow. The angular momentum of a halo changes only through the flux of matter across its boundary; internal radial flows move angular momentum between shells but cannot create it. The rotational kinetic energy changes by the same boundary flux plus a production term, $2\rho u_\varphi^2 u_r/r$, the analogue of Reynolds-stress production in turbulence. For a halo whose mass and radius grow in proportion to time, the azimuthal equation has an exact separable solution, $u_\varphi \propto \sin^{\alpha_\theta}\theta\, M(<r)/r$. It predicts (i) a specific angular momentum of spherical shells proportional to the enclosed mass, $j \propto M^s$ with $s = 1$; (ii) a solid-body core; (iii) a halo angular momentum $H_h = \gamma_H H r_h^2$ with $\gamma_H = 0.43$–$0.51$, i.e. a spin parameter $\lambda' = 0.033$–$0.038$ that does not depend on mass or time. We test these predictions on $2.0 \times 10^6$ haloes of an $N$-body simulation at $z = 0$–$1$, using for every halo a non-rotating counterpart to measure the spin produced by the finite number of particles. This sampling noise, $\lambda'_{\rm noise} \approx 0.31\, n_p^{-1/2}$, is more than half of the measured spin for haloes with fewer than about 150 particles. Haloes resolved with more particles give $s = 0.92$–$1.11$, $\gamma_H = 0.49$–$0.66$ and $\lambda' = 0.037$–$0.050$, close to the solution. Rotation carries only 0.9–1.5 per cent of the kinetic energy and is concentrated to the equatorial plane ($\alpha_\theta \approx 1.1$–$1.5$). The radial velocity anisotropy is the same in the non-rotating counterparts, so it is not induced by spin. Because the mean radial flow of simulated haloes is inward at all resolved radii, rotation is spun up throughout the halo by contraction. At late times, when the concentration grows at a fixed inner structure, the Peebles spin parameter increases while the Bullock spin parameter decreases.

**Key words:** dark matter – galaxies: haloes – methods: numerical – methods: analytical

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## 1 INTRODUCTION

Dark matter haloes rotate, and their angular momentum is one of the main links between the dark matter and the galaxies that form inside it. Gas that cools in a halo keeps a specific angular momentum similar to that of the dark matter, so the spin of the halo sets the sizes of galaxy discs [1, 2]. Observed galaxy sizes indeed scale nearly linearly with the virial radii of their haloes, as this picture predicts [3, 4], and the specific angular momentum of stars follows a tight relation with stellar mass, with the slope expected from the haloes and a normalization that depends on galaxy morphology [5, 6]; in hydrodynamic simulations the spin of a halo also correlates with the morphology of its central galaxy [7]. The direction of halo spin is correlated with the surrounding cosmic web [8], and the resulting intrinsic alignments of galaxies are a main systematic effect in weak-lensing cosmology [9, 10].

The origin of this angular momentum is described by tidal torque theory: a proto-halo gains angular momentum from the misalignment of its inertia tensor with the surrounding tidal field, and

⋆ E-mail: zhijie.xu@pnnl.gov; zhijiexu@hotmail.com

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the angular momentum grows linearly with time until turnaround [11, 12, 13, 14]. Haloes keep growing after turnaround, however, and simulations show that accreted satellites and smooth accretion keep changing their spin [15, 16]. The resulting dimensionless spin $\lambda' = J/(\sqrt{2}MVR)$ has a lognormal distribution with median 0.035 and little dependence on mass or redshift [17, 18, 19, 20], but a derivation of this value from the dynamics of a growing halo is still missing. Inside haloes, the specific angular momentum $j$ of spherical shells grows with enclosed mass roughly as $j \propto M^s$ with $s = 1.3 \pm 0.3$ [18]; discs that inherit this distribution contain too much low-angular-momentum material compared with observed dwarf and disc galaxies, one form of the angular momentum problem of galaxy formation [21, 22]. Measurements of spin are also limited by resolution: with $n_p$ particles, random velocities alone give a halo a non-zero angular momentum. This bias is usually removed by discarding haloes with fewer than a few hundred particles [19], but its size as a function of $n_p$ is rarely stated.

Rotation is not the only departure of haloes from isotropic equilibrium. Haloes are radially anisotropic in their outskirts, and the anisotropy is correlated with the local slope of the density profile [23]. Because Jeans models of galaxies and clusters cannot separate anisotropy from mass, the mass–anisotropy degeneracy [24, 25], it matters whether anisotropy is tied to rotation or to the radial infall of matter. Finally, when accretion slows down, the inner structure of a halo freezes while its virial radius keeps growing with the falling background density, so the concentration grows [26, 27, 28]; how the moment of inertia and the spin parameters respond to this late stage has not been worked out.

In fluid turbulence, the corresponding questions are answered by the equations for the mean flow: the Reynolds stress drains energy from the mean flow into the fluctuations, and vortex stretching passes it to smaller scales [29, 30]. Dark matter haloes are the building blocks of a similar cascade in collisionless dark matter, in which mass and energy are passed from small to large haloes [31, 32, 33], and the growth of a halo fixes a mean radial flow inside it [34]. In this paper we add rotation. We write the moment equations of an axisymmetric halo that rotates and grows, derive exact budget laws for its angular momentum and energies, find an exact self-similar solution for its rotation, and test the predictions on a million haloes of an $N$-body simulation, using for every halo a non-rotating counterpart to measure the spin produced by sampling noise. We find that the angular momentum of a halo changes only through the boundary flux of accreted matter, while its rotational energy changes by the same flux plus a production term analogous to Reynolds-stress production. A halo growing self-similarly rotates with $u_\varphi \propto \sin^{\alpha_\theta}\theta\, M(<r)/r$, which gives $j \propto M$ in spherical shells, a solid-body core and a spin parameter $\lambda' \approx 0.033$–$0.038$ independent of mass and time. In the simulation, sampling noise is more than half of the measured spin for haloes with fewer than about 150 particles; resolved haloes follow the solution to within about 30 per cent, with $s = 0.92$–$1.11$ and $\lambda' = 0.037$–$0.050$. Rotation is a small, equatorially concentrated part of the motion, the radial anisotropy of haloes is not caused by their spin, and rotation is spun up at all radii by the measured inflow. When the concentration grows at a fixed inner structure, the Peebles spin parameter grows by up to a factor 3.6 while the Bullock spin parameter falls.

The paper is organized as follows. Section 2 describes the simulation and the halo samples. Section 3 writes the moment equations of a rotating, growing halo, and Section 4 derives the exact budgets of its momentum and energy. Section 5 presents the self-similar solution, and Section 6 tests it with the simulated haloes. Section 7 treats the late stage, in which the concentration grows at a fixed core. Section 8 discusses the results and Section 9 concludes.

**Table 1.** Parameters of the $N$-body simulation.

| Run | $\Omega_0$ | $\Lambda$ | $h$ | $\Gamma$ | $\sigma_8$ | $L$ (Mpc/$h$) | $N$ | $m_p$ ($M_\odot/h$) | $l_{\rm soft}$ (kpc/$h$) |
|---|---|---|---|---|---|---|---|---|---|
| SCDM1 | 1.0 | 0.0 | 0.5 | 0.5 | 0.51 | 239.5 | $256^3$ | $2.27\times10^{11}$ | 36 |

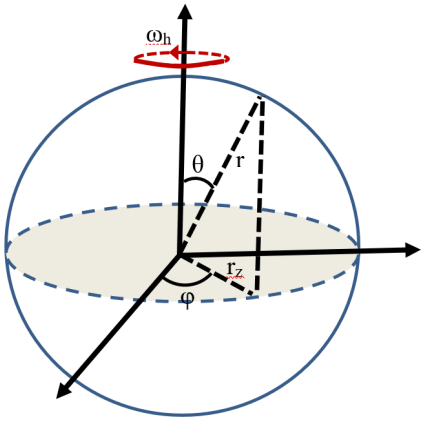



**Figure 1.** Coordinates of a rotating halo: radius $r$, polar angle $\theta$ measured from the rotation axis (the direction of the halo angular momentum), azimuthal angle $\varphi$ and distance to the axis $r_z = r\sin\theta$.

## 2 SIMULATION AND HALO SAMPLES

We use the SCDM run of the Virgo consortium [35, 36] (Table 1). Haloes are identified with a friends-of-friends (FOF) algorithm with a comoving linking length of 0.2 Mpc/$h$ (0.214 times the mean interparticle separation). All haloes with $n_p \geqslant 3$ particles (up to 6326 at $z = 0$) are used at $z = 0$, 0.5, 1 and 2: 718 107 haloes at $z = 0$. The halo data are public [37, 38].

**Frame of each halo.** For every halo we compute the centre of mass, the velocities relative to the halo mean velocity and the angular momentum $\mathbf{H} = \sum_i \mathbf{r}_i \times \mathbf{v}_i$. The direction of $\mathbf{H}$ is the rotation axis (Fig. 1). Each particle has a radius $r$, a polar angle $\theta$ and velocity components $u_r$ (radial, peculiar), $u_\varphi$ (azimuthal, positive in the sense of rotation) and $u_\theta$. Distances are comoving; at $z = 0$ they are also physical.

**Halo size and spin.** The radius $r_h$ of a halo is the outermost radius within which the mean density is $18\pi^2 \approx 178$ times the mean density of the universe (counted from the particle radii). It is undefined for 1 145 haloes at $z = 0$, whose mean density stays below this value at every radius; the other 716 962 haloes enter the statistics. The specific angular momentum is $H_h = |\mathbf{H}|/n_p$ (all particles of the FOF halo), and $m_h$ is the mass within $r_h$. We use the spin parameter of Bullock *et al.* [18],

$$\lambda' = \frac{H_h}{\sqrt{2}V_h r_h}, \qquad V_h^2 = \frac{Gm_h}{r_h}, \tag{1}$$

and the dimensionless angular momentum $\gamma_H = H_h/(Hr_h^2)$, where $H$ is the Hubble parameter. With a mean density of $18\pi^2$ times the background, $V_h = 3\pi H r_h$ and therefore $\lambda' = \gamma_H/(3\pi\sqrt{2})$ exactly.

**A non-rotating counterpart of every halo.** A halo of $n_p$ particles with random velocities has a non-zero angular momentum, and its rotation axis follows that random angular momentum. To measure this effect we build, for every halo, a counterpart without coherent rotation: each particle keeps its radius, its radial velocity and its tangential speed, but is placed in a random direction and given a random tangential direction; positions and velocities are re-centred, and the angular momentum, axis and velocity components are recomputed. The counterpart ('null halo' below) has the same particle number, radial structure, radial anisotropy and kinetic energy as the halo, and an isotropic shape. Any spin it shows is sampling noise.

**Composite haloes.** Profiles are stacked over all haloes in a range of $n_p$ after aligning each halo on its own rotation axis: $n_p = 100$–$300$ ($r_h \approx 0.51\,\mathrm{Mpc}/h$ at $z = 0$), 300–1000 ($r_h \approx 0.74$) and $\geqslant 1000$ ($r_h \approx 1.08$). Means and dispersions are taken over all particles of a bin in $r$ (and, where stated, in $|\cos\theta|$).

## 3 MOMENT EQUATIONS OF A ROTATING, GROWING HALO

Consider a halo that is axisymmetric about its rotation axis, with a spherical density $\rho(r,t)$ and potential. Its mean flow has a radial part $u_r(r,t)$ and an azimuthal part $u_\varphi(r,\theta,t)$; the meridional flow $u_\theta$ is neglected, and the velocity dispersion tensor is diagonal with components $\sigma_{rr}^2$, $\sigma_{\theta\theta}^2$ and $\sigma_{\varphi\varphi}^2$. The continuity equation reduces to

$$\frac{\partial\rho}{\partial t} + \frac{1}{r^2}\frac{\partial(r^2\rho u_r)}{\partial r} = 0, \tag{2}$$

so the azimuthal flow moves matter within spherical shells without compressing it. The radial and azimuthal momentum (Jeans) equations become

$$\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{1}{\rho}\frac{\partial(\rho\sigma_{rr}^2)}{\partial r} + \frac{2\beta_1\sigma_{rr}^2}{r} + \frac{Gm(<r)}{r^2} = 0, \tag{3}$$

$$\frac{\partial u_\varphi}{\partial t} + u_r\frac{\partial u_\varphi}{\partial r} + \frac{u_r u_\varphi}{r} = 0, \tag{4}$$

with the anisotropy parameter that includes the mean rotation,

$$\beta_1 = 1 - \frac{\sigma_{\theta\theta}^2 + \sigma_{\varphi\varphi}^2 + u_\varphi^2}{2\sigma_{rr}^2}, \qquad \beta = 1 - \frac{\sigma_{\theta\theta}^2 + \sigma_{\varphi\varphi}^2}{2\sigma_{rr}^2}. \tag{5}$$

The usual $\beta$ counts rotation as tangential random motion; $\beta_1$ does not. Equation (4) states that the specific angular momentum $r\sin\theta\, u_\varphi$ of a fluid element is conserved as it moves radially: inflow spins matter up, outflow spins it down.

The polar equation adds a relation between $u_\varphi$ and the tangential dispersions, $u_\varphi^2 = \sigma_{\theta\theta}^2 - \sigma_{\varphi\varphi}^2 + \tan\theta\,\partial\sigma_{\theta\theta}^2/\partial\theta$, but only for a spherical density and potential. Simulated haloes are flattened perpendicular to their spin: in the composite haloes with $n_p = 300$–$1000$ the density near the equatorial plane ($|\cos\theta| < 0.25$) is 1.80 times that near the poles ($|\cos\theta| > 0.75$), between $0.3r_h$ and $r_h$. The polar pressure gradients that balance this flattening are an order of magnitude larger than $u_\varphi^2$ (which is 0.07 of $\sigma^2$ at the equator), so the polar equation describes the shape of the halo rather than its rotation, and we do not use it.

## 4 EXACT BUDGETS OF MOMENTUM AND ENERGY

Integrating the moment equations over a halo of radius $r_h(t)$, and keeping the terms that arise because the boundary moves, gives exact budgets. Let $\langle f\rangle_k(r) = \int_0^\pi f\sin^k\theta\,\mathrm{d}\theta$.

**Angular momentum.** Multiplying Eq. (4) by $\rho$, adding $u_\varphi$ times Eq. (2), multiplying by $r\sin\theta$ and integrating over the halo, the internal terms cancel and

$$\frac{\mathrm{d}\bar{H}_h}{\mathrm{d}t} = 2\pi r_h^3\rho(r_h)\,\langle u_\varphi\rangle_2(r_h)\left[\frac{\mathrm{d}r_h}{\mathrm{d}t} - u_r(r_h)\right], \tag{6}$$

where $\bar{H}_h = \int 2\pi r^3\rho\langle u_\varphi\rangle_2\mathrm{d}r$ is the total angular momentum. The bracket is the speed at which matter crosses the moving boundary, so Eq. (6) is the angular momentum carried in by accretion. Radial flows inside the halo redistribute angular momentum between shells but cannot change the total. A halo gains angular momentum if the accreted matter rotates in the same sense as the halo, and loses it otherwise; this is the accretion channel of spin growth seen in simulations [15, 16], which acts after tidal torques have stopped.

**Rotational kinetic energy.** Combining Eq. (4) with Eq. (2),

$$\frac{\partial(\rho u_\varphi^2)}{\partial t} + \frac{1}{r^2}\frac{\partial(r^2\rho u_\varphi^2 u_r)}{\partial r} + \underbrace{2\rho u_\varphi^2\frac{u_r}{r}}_{P_{u\varphi}} = 0, \tag{7}$$

and, for $\bar{K}_a = \frac{1}{2}\int 2\pi r^2\rho\langle u_\varphi^2\rangle_1\mathrm{d}r$,

$$\frac{\mathrm{d}\bar{K}_a}{\mathrm{d}t} = \pi r_h^2\rho(r_h)\langle u_\varphi^2\rangle_1(r_h)\left[\frac{\mathrm{d}r_h}{\mathrm{d}t} - u_r(r_h)\right] - \int_0^{r_h} 2\pi r\rho u_r\langle u_\varphi^2\rangle_1\mathrm{d}r. \tag{8}$$

The first term is again the boundary flux. The second is a production term: the stress $\rho u_\varphi^2$ acting on the radial strain rate $u_r/r$. It plays the role of the Reynolds-stress production that feeds turbulence from the mean flow; here it takes energy from the rotation where matter moves out ($u_r > 0$) and gives energy to it where matter falls in ($u_r < 0$). Unlike in turbulence, the exchange can go either way.

**Radial momentum, radial energy and the virial balance.** The same steps applied to Eq. (3) give the radial momentum and radial kinetic energy budgets and a virial theorem for a rotating, growing halo,

$$\frac{\mathrm{d}\bar{G}_h}{\mathrm{d}t} + S_h - 2\bar{K}_r - 2\bar{K}_a - 2\bar{K}_\sigma - \bar{W} = 0, \tag{9}$$

where $\bar{G}_h = \int 4\pi r^3\rho u_r\mathrm{d}r$, $S_h = 4\pi r_h^3\rho(r_h)[u_r(r_h)(u_r(r_h) - \mathrm{d}r_h/\mathrm{d}t) + \frac{1}{2}\langle\sigma_{rr}^2\rangle_1(r_h)]$ collects the surface terms, $\bar{K}_r$, $\bar{K}_a$ and $\bar{K}_\sigma$ are the kinetic energies of radial flow, rotation and random motion, and $\bar{W} = -\int 4\pi r^3\rho\,\partial_r\phi\,\mathrm{d}r$ is the potential energy. Rotation enters the virial balance exactly as random kinetic energy does.

## 5 A SELF-SIMILAR SOLUTION FOR ROTATING, GROWING HALOES

### 5.1 Radial flow

Write the enclosed mass as $m(<r,t) = m_h(t)F(x)/F(c)$ with $x = r/r_s(t)$ and concentration $c = r_h/r_s$. The continuity equation then gives the mean radial flow [34]

$$u_r = \frac{r_s}{t}u_h(x), \qquad u_h = x\frac{\partial\ln r_s}{\partial\ln t} + \left(\frac{\partial\ln F(c)}{\partial\ln t} - \frac{\partial\ln m_h}{\partial\ln t}\right)\frac{F(x)}{F'(x)}. \tag{10}$$

Two limits are useful. A halo in its early, fast-accretion stage, with constant concentration and $m_h \propto r_h \propto r_s \propto t$ (so that its mean density stays $18\pi^2$ times the background), has $u_h = x - F/F'$. A halo in its late stage, with fixed $r_s$ and core mass and $m_h \propto F(c)$, has $u_h = 0$.

### 5.2 Azimuthal flow

We look for a separable rotation, $u_\varphi = \omega_h(t)\, r_s(t)\, F_\varphi(x) \sin^{\alpha_\theta}\theta$, where $\omega_h$ is the angular velocity of the halo. Substituting into Eq. (4) with the early-stage flow gives $\partial \ln F_\varphi / \partial \ln x = (2 + \partial \ln \omega_h / \partial \ln t)\, \partial \ln F / \partial \ln x - 1$. If the angular velocity scales with the expansion rate, $\omega_h \propto H \propto t^{-1}$, the exact solution is

$$u_\varphi = \alpha_f\, \omega_h r_s \sin^{\alpha_\theta}\theta\, \frac{F(x)}{x} \;\propto\; \sin^{\alpha_\theta}\theta\, \frac{Gm(<r)}{r}. \tag{11}$$

The rotation speed is proportional to the square of the circular velocity. Three consequences follow directly.

(i) The specific angular momentum of a spherical shell, $j(r) = r\langle \sin\theta\, u_\varphi\rangle$, is proportional to the mass inside it: $j \propto M(<r)^s$ with $s = 1$.

(ii) The angular velocity $u_\varphi / r_z \propto Gm(<r)/r^2$ follows the gravitational acceleration. For an NFW profile ($F = \ln(1+x) - x/(1+x)$, Navarro *et al.* [39]) $F \propto x^2$ at small $x$, so the core rotates as a solid body and the outer halo rotates more slowly.

(iii) Angular momentum is conserved along radial trajectories (Eq. 4), so the solution needs no assumption about the source of the angular momentum other than its time dependence.

The constant $\alpha_f$ follows from equating the angular momentum of Eq. (11) with $H_h = \frac{2}{3}\omega_h r_g^2$, where $r_g$ is the rms radius:

$$\alpha_f = \frac{8c^2}{3F(c)}\, \frac{r_g^2}{r_h^2}\, \frac{\Gamma(2+\alpha_\theta/2)}{\sqrt{\pi}\,\Gamma(3/2+\alpha_\theta/2)}. \tag{12}$$

### 5.3 Closure at the boundary and the spin parameter

One condition fixes $\omega_h$. We assume that matter arriving at the equator of the halo has equal azimuthal and radial peculiar speeds, $u_\varphi(r_h, \pi/2) = -u_{rp}(r_h)$, where $u_{rp} = u_r - Hr = (r_s/t)(x/3 - F/F')$ is the peculiar radial flow. This gives

$$\omega_h = \Big(\frac{3}{2\alpha_h} - \frac{1}{2}\Big)\, \frac{c^2}{F(c)\,\alpha_f}\, H, \qquad \alpha_h = \frac{cF'(c)}{F(c)}, \tag{13}$$

so a halo in the early stage rotates at a fixed multiple of the expansion rate, and its angular momentum is

$$H_h = \gamma_H H r_h^2, \qquad \gamma_H = \frac{1}{8}\Big(\frac{3}{\alpha_h} - 1\Big)\, \frac{\sqrt{\pi}\,\Gamma(3/2+\alpha_\theta/2)}{\Gamma(2+\alpha_\theta/2)}. \tag{14}$$

Since $r_h \propto t$, the specific angular momentum grows as $H_h \propto t$ and the total as $\bar{H}_h \propto t^2$, all of it supplied by the boundary flux of Eq. (6), while the specific rotational energy stays constant. The spin parameters are

$$\lambda' = \frac{\gamma_H}{3\pi\sqrt{2}}, \qquad \lambda_p = \frac{H_h |E_h|^{1/2}}{G m_h^{5/2}} = \frac{\gamma_H}{3\pi}\sqrt{\gamma_\Phi\big(1 + \tfrac{n_e}{2}\big)}, \tag{15}$$

where $\Phi_h = -\gamma_\Phi G m_h / r_h$ is the specific potential energy and $n_e = 2K_h/\Phi_h$ the effective virial exponent of a growing halo, which includes its surface energy [34]. Neither depends on mass or time. Table 2 lists the values for an NFW halo with $c = 3.5$, the concentration at which a self-similarly growing NFW halo has no net radial momentum [34], and for an isothermal halo.

For the NFW halo the radial flow reverses sign at $x = 2.16$: matter moves out inside this radius and in outside it. The specific kinetic energy of the peculiar radial flow is $K_{rp} = 0.194 H^2 r_h^2$, that of rotation $K_a = 0.766 H^2 r_h^2$ ($\alpha_\theta = 0$), and the potential energy $\Phi_h = -8.42\pi^2 H^2 r_h^2$. All specific energies are constant in time; the rotational energy of the whole halo grows as $m_h \propto t$, of which the boundary flux supplies 0.846 and the production term removes 0.080 (units $m_h H^2 r_h^2 / t$).

**Table 2.** The self-similar rotating solution. NFW: $c = 3.5$, $\alpha_h = 0.833$, $r_g^2/r_h^2 = 0.321$, $\gamma_\Phi = 0.936$, $n_e = -1.42$. Isothermal: $F = x/c$, $\alpha_h = 1$, $r_g^2/r_h^2 = 1/3$, $\gamma_\Phi = 1$, $n_e = -3/2$. Energies in units of $H^2 r_h^2$.

| | NFW, $\alpha_\theta = 0$ | NFW, $\alpha_\theta = 1$ | Iso., $\alpha_\theta = 0$ | Iso., $\alpha_\theta = 1$ |
|---|---|---|---|---|
| $\omega_h/H$ | 2.38 | 2.02 | 1.77 | 1.50 |
| $\gamma_H$ | 0.51 | 0.43 | 0.393 | 0.333 |
| $K_a$ | 0.766 | 0.511 | 0.500 | 0.333 |
| $\lambda'$ | 0.038 | 0.033 | 0.029 | 0.025 |
| $\lambda_p$ | 0.028 | 0.024 | 0.021 | 0.018 |

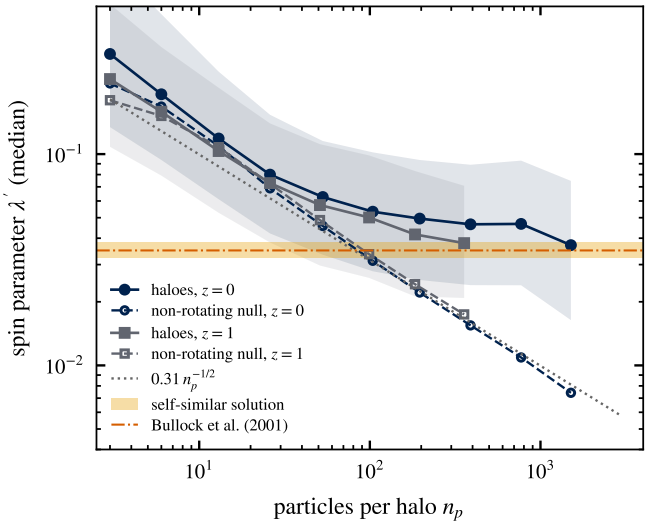



**Figure 2.** Median spin parameter $\lambda'$ against particle number for haloes (solid; shaded: 16–84 per cent range) and for their non-rotating counterparts (dashed), at $z = 0$ and $z = 1$. Dotted: $\lambda' = 0.31 n_p^{-1/2}$. Band: the self-similar solution ($\alpha_\theta = 0$–1); dash-dotted: median of Bullock *et al.* [18]. Below $n_p \approx 150$ the spin is dominated by sampling noise.

## 6 TESTS WITH SIMULATED HALOES

### 6.1 Sampling noise sets a resolution limit for spin

Fig. 2 shows the median spin parameter of haloes and of their non-rotating counterparts as a function of particle number. The spin of the counterparts falls as $\lambda'_{\rm noise} = 0.31 n_p^{-1/2}$ at all redshifts (fitted coefficient 0.31–0.35 for $z = 0$–2), as expected for the sum of $n_p$ random angular momenta. The haloes follow the counterparts for $n_p \lesssim 40$; their median spin is twice the noise at $n_p \approx 150$ and three times the noise at $n_p \approx 390$ ($z = 0$). Spins, rotation axes and rotation profiles of haloes with fewer than about a hundred particles are therefore mostly sampling noise, and properties of 'small' haloes derived from them describe the noise rather than the haloes. In what follows we use haloes with $n_p \geqslant 100$ and show the counterparts for comparison. For $150 \leqslant n_p < 1200$ the median spin is $\lambda' = 0.047$–$0.050$ at $z = 0$, and 0.037 for $n_p \geqslant 1200$; at $n_p = 300$–$600$ the noise is 0.33 of the median spin.

### 6.2 Rotation profiles

Fig. 3 shows the rotation of composite haloes with $n_p = 300$–$1000$ at $z = 0$ in four bins of polar angle. Rotation is fastest in the equatorial plane and weak near the poles. Fitting $u_\varphi \propto \sin^{\alpha_\theta}\theta$ in each radial

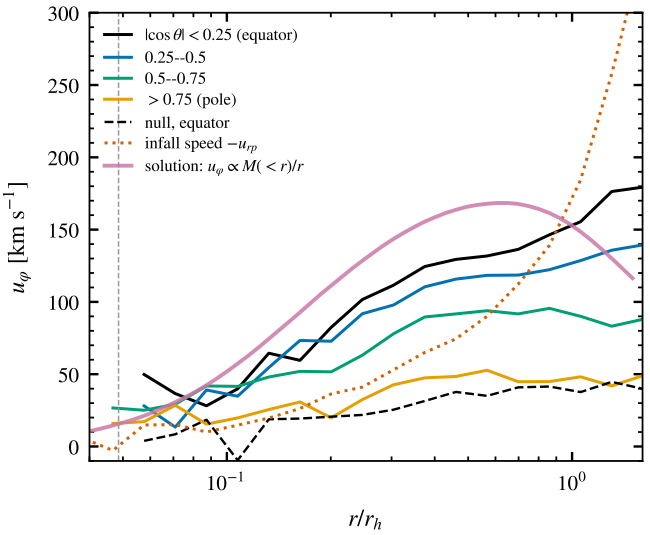



**Figure 3.** Mean azimuthal velocity of composite haloes with $n_p = 300$–$1000$ at $z = 0$ in four bins of polar angle (solid), for the non-rotating counterparts at the equator (dashed), the mean infall speed $-u_{rp}$ (dotted) and the solution $u_\varphi \propto M(<r)/r$ with the measured mass profile, normalised at $r_h$ (pink). Vertical line: softening length.

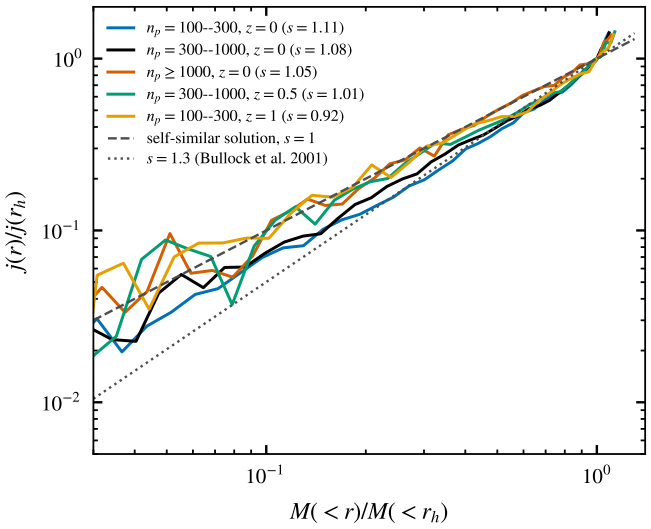



**Figure 4.** Specific angular momentum of spherical shells against enclosed mass, both normalised at $r_h$, for composite haloes at $z = 0$, 0.5 and 1. Dashed: the self-similar solution, $s = 1$; dotted: $s = 1.3$.

bin between $0.2r_h$ and $r_h$ gives a median $\alpha_\theta = 1.48$ (1.38 and 1.06 for $n_p = 100$–$300$ and $\geqslant 1000$): the rotation is roughly cylindrical, with $\alpha_\theta$ between the values 0 and 1 of Table 2 and slightly above. Rotation is a small part of the motion: inside $r_h$ the mean azimuthal flow carries 0.9–1.5 per cent of the kinetic energy (0.3 per cent or less in the counterparts), and even at the equator $u_\varphi^2$ is only 0.05–0.09 of the dispersion $\sigma^2$, i.e. $u_\varphi/\sigma \approx 0.22$–$0.30$.

The equatorial rotation speed rises outwards. With the measured mass profile, the solution $u_\varphi \propto M(<r)/r$ (Eq. 11) reproduces the rise inside $0.5r_h$ to within about 30 per cent; beyond $0.6r_h$ the measured rotation keeps rising while $M(<r)/r$ falls. In this outer region the infall speed exceeds the rotation speed and the halo is not in the steady self-similar state assumed by the solution.

### 6.3 The angular momentum profile

Fig. 4 shows the specific angular momentum of spherical shells against the enclosed mass. Between 5 per cent of the mass and $r_h$ the profiles are power laws with $s = 1.11$, 1.08 and 1.05 for $n_p = 100$–$300$, 300–1000 and $\geqslant 1000$ at $z = 0$, and $s = 1.01$ and 0.92 at $z = 0.5$ and 1. Subtracting the angular momentum of the non-rotating counterparts changes $s$ by less than 0.12. The solution predicts $s = 1$; the measured profiles are close to it and at the lower end of the range $s = 1.3 \pm 0.3$ found for well-resolved haloes by Bullock *et al.* [18]. Within the self-similar picture a linear $j(M)$ is the signature of angular momentum that is conserved along radial trajectories while mass and size grow together; it requires no particular history of tidal torques.

### 6.4 Spin against mass and redshift

Fig. 5a shows $\gamma_H = H_h/(Hr_h^2)$ for haloes at $z = 0$, 0.5 and 1. For resolved haloes ($n_p \geqslant 150$) the medians are 0.49–0.66, compared with 0.43–0.51 for the solution; the corresponding spin parameters are $\lambda' = 0.037$–$0.050$ against 0.033–0.038. The dependence on mass is weak, as predicted, and the values increase slowly with time: for $n_p = 300$–$600$, $\gamma_H = 0.51$, 0.59 and 0.62 at $z = 1$, 0.5 and 0. The angular velocity, $\omega_h = H_h/\langle r_z^2 \rangle \approx \frac{3}{2}\gamma_H H r_h^2/r_g^2$, has medians $\omega_h/H = 2.0$ at $z = 0$ and 1.3 at $z = 1$; it is lower at higher redshift mainly because the FOF haloes are more extended there ($r_g^2/r_h^2 = 0.38$ and 0.48).

The boundary closure is tested in Fig. 5b, which compares the equatorial rotation speed at $r_h$ with the infall speed there. The two are equal to within 30 per cent for $n_p = 100$–$1000$ at $z = 0$ (ratios 1.29 and 0.92), but the infall speed grows faster than the rotation for more massive haloes (0.61) and at higher redshift (0.56 and 0.37 at $z = 1$), i.e. for haloes that accrete fastest. The closure therefore holds on average for typical haloes today, and the angular momentum coefficient $\gamma_H$, which depends on it, agrees with the measurements to within about 30 per cent at all redshifts studied.

### 6.5 Anisotropy is not induced by spin

Rotation could make the dispersions anisotropic: matter far from the rotation axis might be hotter than matter near it. Fig. 6a tests this for haloes with $n_p = 500$–$1000$. The dispersion of the 1690 particles within 0.016 Mpc/$h$ of the rotation axis, pooled over all haloes, is 1.03 times the dispersion of all particles at the same radius (median over $0.05 < r < 1$ Mpc/$h$; 68 per cent range 0.83–1.14). Velocity dispersion does not depend on the distance to the rotation axis. An 'axial dispersion' far below the shell dispersion is obtained only if the variance is computed separately in every group of haloes with the same $n_p$ and then averaged: 96 per cent of these groups contain a single particle near the axis, whose variance is zero, and the average falls to 2.6 per cent of the true value.

Fig. 6b shows the anisotropy $\beta(r)$ of the composite haloes and of their non-rotating counterparts. $\beta$ grows from about zero in the centre to 0.3–0.4 near $r_h$, in the same way in both: at $r_h/2$, $\beta = 0.27$ for the haloes and 0.25 for the counterparts ($n_p = 300$–$1000$). Since $u_\varphi^2$ is a few per cent of $\sigma^2$, the shell-averaged $\beta_1$ and $\beta$ of Eq. (5) differ by less than 0.03. The radial anisotropy of haloes comes from the radial structure of their orbits, i.e. from infall, and not from rotation.

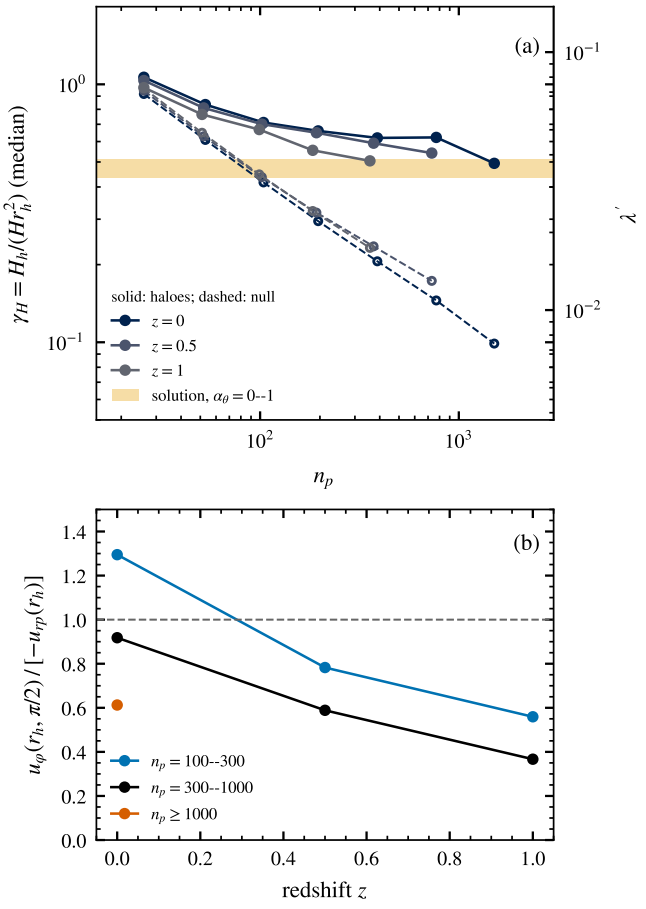



**Figure 5.** (a) Median $\gamma_H = H_h/(Hr_h^2)$ (right axis: $\lambda' = \gamma_H/3\pi\sqrt{2}$) against $n_p$ at $z = 0$, 0.5 and 1, for haloes (solid) and their non-rotating counterparts (dashed). Band: the solution, $\alpha_\theta = 0$–1. (b) Ratio of the equatorial rotation speed to the infall speed at $r_h$ (the closure assumes 1).

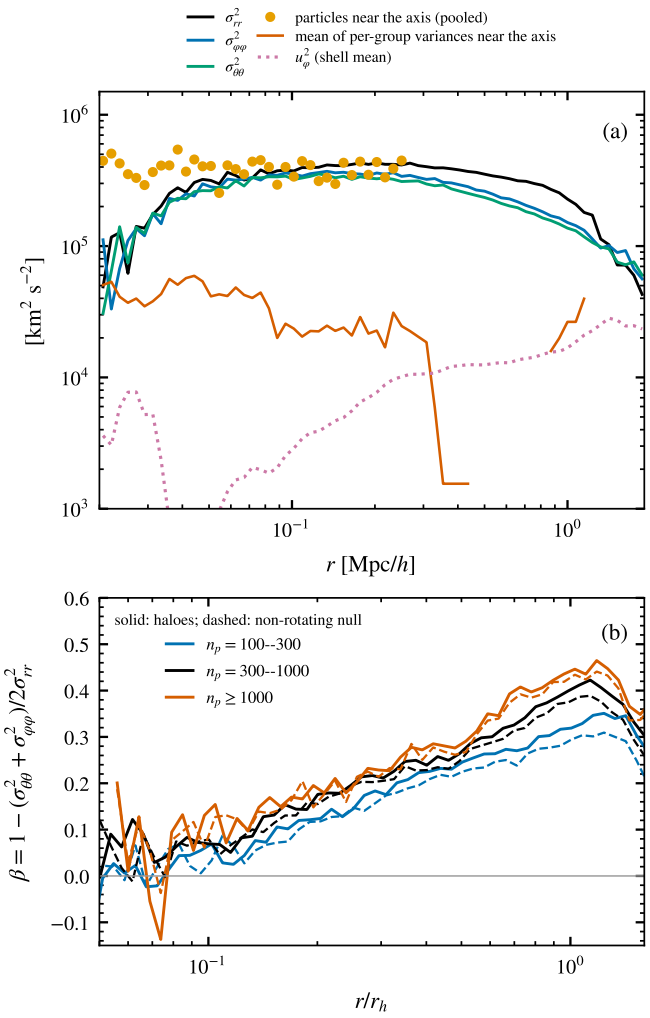



**Figure 6.** (a) Composite haloes with $n_p = 500$–1000 at $z = 0$: dispersions in spherical shells, dispersion of the particles near the rotation axis pooled over all haloes (dots), the average of the per-group variances near the axis (red), and $u^2_\varphi$. (b) Anisotropy $\beta(r)$ of composite haloes (solid) and their non-rotating counterparts (dashed).

For Jeans modelling this means that the anisotropy profile cannot be inferred from the spin, and that rotation itself, at the level of a few per cent of the kinetic energy, can be neglected.

### 6.6 Radial flow and the exchange of energy

For the self-similar NFW halo the radial flow is outward inside $x = 2.16$, so the production term in Eq. (7) takes energy from rotation in the core and gives it to rotation outside (Fig. 7a); the radial flow itself loses energy to random motion for $x < 1$, gains between $x = 1$ and 2.16, and loses beyond. The simulated haloes do not show the outward flow. Their mean radial velocity, including the Hubble flow, is negative in 92–100 per cent of the radial bins between twice the softening length and $r_h$ (Fig. 7b; see also Xu [34]). With $u_r < 0$ everywhere, the production term is negative everywhere: rotational energy is created at all radii as matter falls in and conserves its angular momentum, and the halo is spun up from the inside out. The two-way exchange predicted for the core is a property of the idealised self-similar flow, not of the simulated haloes.

## 7 THE LATE STAGE: GROWTH OF CONCENTRATION AT A FIXED CORE

When accretion slows, the scale radius and the mass within it stop changing while the concentration grows; the radial flow vanishes ($u_h = 0$ in Eq. 10) and the halo mass grows as $m_h \propto F(c)$. To describe the change of shape we use an Einasto profile, $F(x, \alpha) = \gamma(3/\alpha, 2x^\alpha/\alpha)$, and the core mass fraction $C_F = F(1, \alpha)/F(c, \alpha)$. An early-stage halo with $c = 3.5$ and $\alpha = 0.2$ has $C_F = 0.27$.

Fig. 8 shows paths in the $(c, \alpha)$ plane. Path 1 keeps $C_F$ and the halo mass fixed; it ends at $\alpha \approx 0.71$. Path 2 follows the relations $\alpha(\nu)$ and $c(\nu)$ measured for simulated haloes as a function of peak height [40]: from $\nu = 5$ to 0.5, $\alpha$ falls to 0.119, $c$ rises to 20.7 and $C_F$ falls to 0.075. Path 3 is a simplified version, with fixed $\alpha = 0.2$ after the early stage; along it $C_F$ falls to 0.083.

Along path 3 the inner density stays fixed while the outer profile extends (Fig. 9a), and the moment of inertia $F_\omega = r_g^2/r_s^2$ grows from 3.94 at $c = 3.5$ to 712, while for an NFW profile it diverges (Fig. 9b). This growth of concentration at a fixed core is the pseudo-evolution

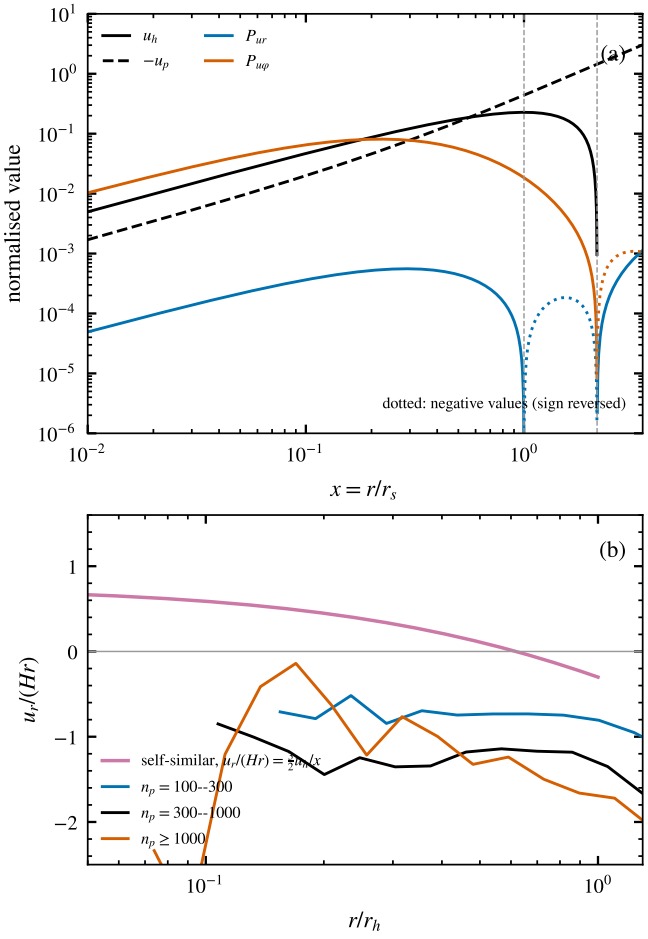



**Figure 7.** (a) Self-similar NFW halo ($c = 3.5$): radial flow $u_h$, peculiar radial flow $u_p$ and the production terms of radial ($P_{ur}$) and rotational ($P_{u\varphi}$) kinetic energy, in units of $2\bar{\rho}_h v_{\rm cir}^2/t$; positive production removes energy from the mean flow. (b) Measured mean radial velocity (Hubble flow included) of composite haloes at $z = 0$, against the self-similar prediction.

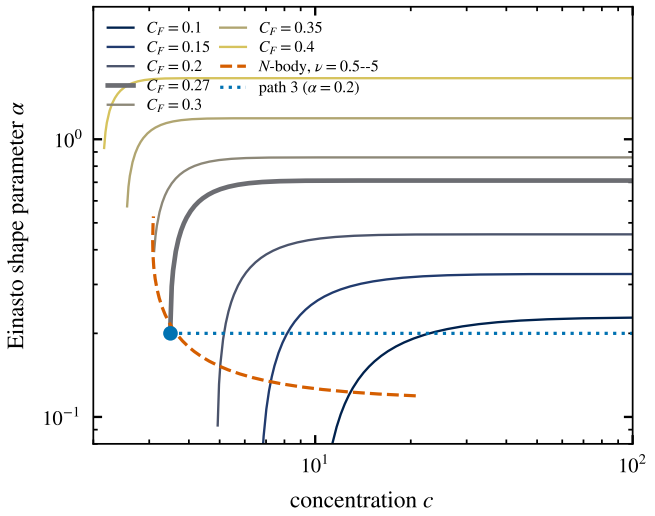



**Figure 8.** Paths of halo evolution in the plane of concentration $c$ and Einasto shape parameter $\alpha$. Solid lines: constant core mass fraction $C_F$ (thick: $C_F = 0.27$, path 1). Dashed: relation of simulated haloes for peak heights $\nu = 0.5$–5 (path 2). Dotted: path 3, fixed $\alpha = 0.2$ after the early stage (dot).

of halo structure [28], here applied to the moment of inertia. Without accretion the angular momentum is fixed (Eq. 6); with slow accretion it grows. If, as in the early stage, the specific rotational energy stays constant, then $H_h \propto r_g$ and $\omega_h \propto r_g^{-1}$. Fig. 10 shows the consequences: the mass grows by a factor 3.23, $r_g$ and $H_h$ by 13.4, $\omega_h$ falls to 0.074 of its initial value and the potential $\Phi_h$ to 0.77. The Peebles spin parameter, which measures angular momentum against the binding energy dominated by the fixed core, grows by a factor 3.6; the Bullock spin parameter, which measures it against the growing virial radius and velocity, first stays near its initial value and then falls to 0.14 of it. Statements about the evolution of 'the' spin parameter at late times therefore depend on its definition. In the simulation, at fixed $n_p = 300$–600 the median $\lambda'$ rises from 0.038 at $z = 1$ to 0.047 at $z = 0$, but these are different haloes at different stages, not the same haloes followed in time.

## 8 DISCUSSION

**What sets the spin of haloes.** The budget law (6) makes the growth of angular momentum after turnaround a boundary process: the accreted matter brings its angular momentum, and internal flows only redistribute it. If the rotation speed of matter arriving at the equator is comparable to its infall speed, a halo that grows in proportion to time has $H_h = \gamma_H H r_h^2$ with $\gamma_H \approx 0.4$–0.5 and $\lambda' \approx 0.03$–0.04, independent of mass and time. The measured spins of resolved haloes are within about 30 per cent of this value (FOF haloes, which include unrelaxed systems, have the highest spins among common halo definitions; Bett *et al.* [19] find medians that differ by 15 per cent between definitions), and the closure itself holds to 30 per cent for typical haloes at $z = 0$. This offers a simple reason why the spin parameter depends so weakly on halo mass and time.

**Internal distribution.** The solution predicts a linear $j(M)$ profile. The measured slopes, $s \approx 0.92$–1.11, are close to 1 and lower than the mean value 1.3 of Bullock *et al.* [18], which was measured for haloes with more than $10^3$ particles in a ΛCDM simulation. A linear $j(M)$ still contains more low-$j$ material than an exponential disc, so the angular momentum problem of disc formation is not removed by halo dynamics alone.

**Anisotropy and rotation.** The radial anisotropy of haloes is identical in haloes and in their non-rotating counterparts, and the dispersion near the rotation axis equals that elsewhere. Rotation is a small (about one per cent) part of the kinetic energy and does not shape the velocity ellipsoid. Halo flattening perpendicular to the spin axis is a much larger effect than rotation in the polar balance.

**Resolution.** Spins of haloes with fewer than about fifty particles are dominated by sampling noise, and the noise is more than half of the spin below about 150 particles, $\lambda'_{\rm noise} \approx 0.31 n_p^{-1/2}$. This simple law can be used to correct spin distributions of poorly resolved haloes, and it implies that the frequently used cut of 300 particles [19] keeps the noise below about a third of the median spin. The isotropic counterparts slightly overestimate the noise of elongated haloes: at $z = 2$, where small haloes are most elongated, their spin exceeds that of the haloes by 15–20 per cent.

**Limitations.** The solution assumes spherical density, axisymmetric rotation and self-similar growth with $\omega_h \propto H$; the simulated haloes are flattened, accrete episodically and have an inward mean radial flow at all radii. FOF haloes include infalling matter near $r_h$, where the closure and the outer rotation profile are least reliable. The

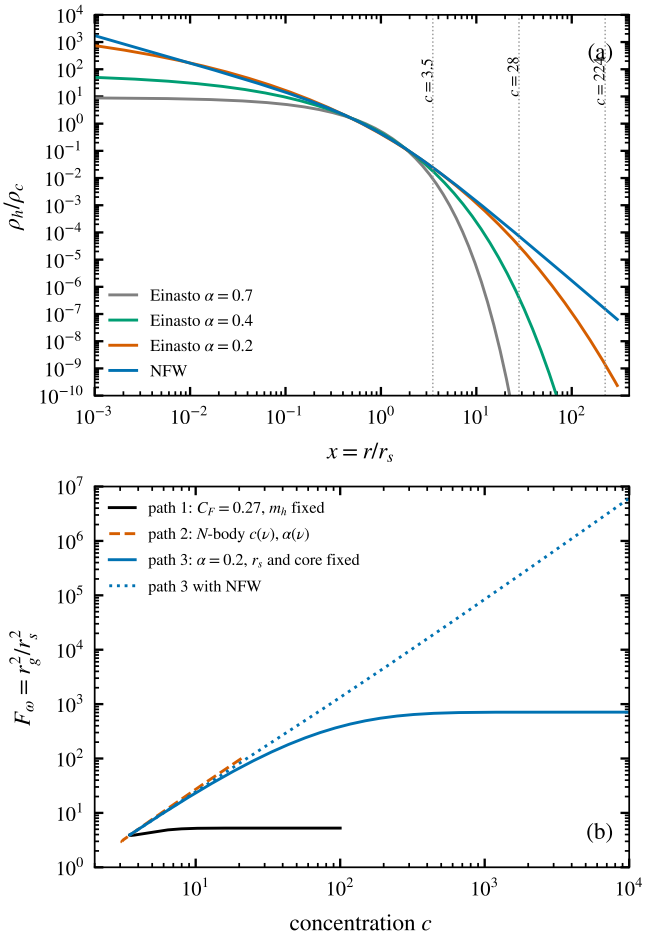



**Figure 9.** (a) Density relative to the mean core density, $\rho_h/\rho_c = F'/(3F(1)x^2)$, for Einasto profiles and NFW; vertical lines mark $c = 3.5$, 28 and 224 along path 3. (b) Moment of inertia $F_\omega = r_g^2/r_s^2$ along the three paths of Fig. 8.

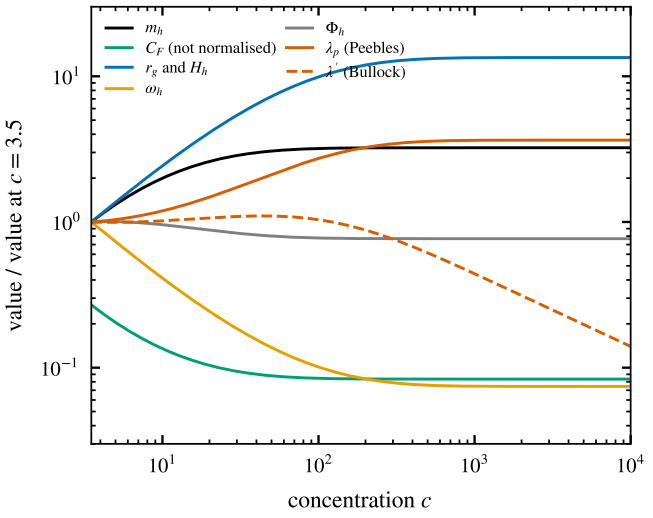



**Figure 10.** Path 3 (Einasto $\alpha = 0.2$, fixed $r_s$ and core mass, constant specific rotational energy): halo mass, core mass fraction, rms radius and angular momentum, angular velocity, potential and the two spin parameters, normalised to their values at $c = 3.5$ (except $C_F$).

simulation is an SCDM run; the scalings with $H$ and $r_h$ should carry over to ΛCDM, but the numbers should be checked. The late-stage evolution assumes a constant specific rotational energy, which is a hypothesis that tracking of individual haloes could test.

## 9 CONCLUSIONS

(i) The angular momentum of a growing halo changes only by the flux of accreted matter across its boundary; its rotational energy changes by the same flux plus a production term $2\rho u_\varphi^2 u_r/r$, the analogue of Reynolds-stress production in turbulence (Eqs. 6–8).

(ii) A halo whose mass and radius grow in proportion to time rotates as $u_\varphi \propto \sin^{\alpha_\theta}\theta\, M(<r)/r$, with $j \propto M$ in spherical shells, a solid-body core, $\omega_h \propto H$ and a spin parameter $\lambda' = 0.033$–$0.038$ ($\lambda_p = 0.024$–$0.028$) independent of mass and time.

(iii) In the simulation, the spin produced by sampling noise is $\lambda'_{\rm noise} = 0.31 n_p^{-1/2}$; it is more than half of the measured spin for haloes with fewer than about 150 particles and comparable to it below about 50.

(iv) Resolved haloes have $j \propto M^s$ with $s = 0.92$–$1.11$, $\gamma_H = 0.49$–$0.66$ and $\lambda' = 0.037$–$0.050$, within about 30 per cent of the solution; the boundary closure holds to 30 per cent for typical haloes at $z = 0$ and underestimates the infall of rapidly accreting haloes.

(v) Rotation carries 0.9–1.5 per cent of the kinetic energy and is concentrated to the equatorial plane ($\alpha_\theta \approx 1.1$–$1.5$); the dispersion near the rotation axis equals the dispersion elsewhere; and the radial anisotropy of haloes is not induced by their spin.

(vi) The mean radial flow of simulated haloes is inward at all resolved radii, so rotation is spun up throughout the halo by contraction.

(vii) When the concentration grows at a fixed core, the Peebles spin parameter grows (by up to 3.6 times) while the Bullock spin parameter falls; the late-time evolution of spin depends on its definition.

## DATA AVAILABILITY

The halo-based statistics of the simulation are available on Zenodo [37, 41] and GitHub [38]. The per-particle data used here, the code that computes every number and figure, and the CSV files of all results are in the reproducible package that accompanies this paper.

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